00-S-355B
July 26, 1971
SUPERSEDING
Fed. Spec. 00-S-355A
April 17, 1967

FEDERAL SPECIFICATION

SIFTING MACHINE, FLOUR, ELECTRIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a commercial flour sifting machine, stand mounted on casters with electric motor.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

N-F-481	-	Flour, Wheat.
CC-M-636	-	Motor, Alternating-Current, (Fractional Horsepower).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 7320

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Ft. Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-T-152	-	Treatment, Moisture-and Fungus-Resistant, of Communications, Electrical, and Associated Electrical Equipment.
MIL-B-2316	-	Bakery Equipment, Including Units, Assemblies, Repair Parts and Tools, Preparation for Delivery Of.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements For Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement Of.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories (UL), Inc. Standard:

No. 73 - Motor-Operated Appliances.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio St., Chicago, Ill., 60611; 1285 Walt Whitman Road, Melville, Long Island, N.Y., 11746; or 1655 Scott Blvd., Santa Clara, Calif., 95050.)

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Baking Industry Sanitation Standards Committee (BISSC) Standard:

No. 1 - Flour Handling Equipment for Bakeries.

(Application for copies should be addressed to Baking Industry Sanitation Standards Committee, 521 Fifth Ave., New York, N.Y. 10017.)

American Society for Testing and Materials (ASTM) Standard:

A 167-63 - Corrosion-Resisting Chromium Steel Plate, Sheet and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article testing and approval (see 4.2 and 6.3).

3.2 Standard product. The flour sifting machines delivered under this specification shall be the manufacturer's commercial standard product, except for any changes necessary to comply with specification requirements.

3.3 Standards compliance.

3.3.1 Baking Industry Sanitary Standards (BISSC). The flour sifting machine shall conform to the applicable requirements of BISSC Std. No. 1.

3.3.1.2 BISSC certification. The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the flour sifting machine he proposes to furnish under this specification conforms to the applicable requirements of BISSC Std. No. 1 as follows:

(1) A photostatic copy of the current BISSC certificate of authorization for the flour sifting machine shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to the approval of the first shipment; or

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(2) A certified test report from a recognized independent testing laboratory acceptable to the medical authority functioning in an advisory capacity to the preparing agency of the specification indicating that the flour sifting machines have been tested and conform to the applicable requirements of BISSC Std. No. 1 shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to the approval of the first shipment.

3.3.2 Underwriters' Laboratories Inc. (UL). The electrical components utilized in the flour sifter shall conform to the applicable requirements of UL Std. No. 73.

3.3.2.1 UL certification. Prior to the approval of the preproduction sample, if one is submitted, or prior to the approval of the first shipment, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that all electric components of flour sifters he proposes to furnish under this specification conform to the applicable requirements of UL Std. No. 73. Acceptable evidence of meeting the requirements of UL Std. No. 73 shall be the UL listing mark, or label; or a certified test report from an independent testing laboratory acceptable to the Government indicating that the electrical components of the flour sifter have tested and conform to all the requirements of UL Std. No. 73.

3.4 Material. Materials not definitely specified shall be of the quality normally used by the manufacturer for his flour sifters provided the completed flour sifters conform to all provisions of this specification.

3.4.1 Corrosion-resisting steel sheet. Corrosion-resisting steel sheet shall conform to the requirements for any type of the 300 series of ASTM A 167-63.

3.5 Design and construction. The flour sifter shown in figure 1 herein is for illustrative purposes only, except for dimensions specified, and is not intended to exclude procurement of flour sifters otherwise meeting all requirements of this specification. The sifting machine shall consist essentially of a flour hopper, 14-mesh sifting screen, an electric motor driven sifting mechanism mounted on top of stand, a discharge spout, an electric cord, and a stand mounted on 4 casters. The sifting machine with the stand removed shall mount on a bench. The base of the machine shall be provided with a 3/8-inch diameter hole in each corner. Four corresponding 3/8-inch diameter holes for mounting the sifter shall be provided in the horizontal surface of the upper angle iron frame girt of the stand. A discharge spout shall be provided at the base of the hopper and the lowest point of the spout shall be 40 to 42 inches above the floor level. When

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tested as specified in 4.4.1, the machine shall sift not less than 40 pounds of white flour per minute through a standard 14-mesh screen. When tested as specified in 4.4.2, the sifting machine shall operate without evidence of over heating, excessive vibration, and the moving components shall not bind or function improperly.

3.5.1 Stand. The stand shall be of such construction that members may be disassembled for shipping. A step shall be attached to the stand in such a manner as to permit the step to be f. .ded back when not in use and to be contained within the stand. When tested as specified in 4.4.3, the step shall support a 400-pound load without any resulting permanent set to the step tread or failure of any connecting part and the stand shall not tilt or shift under the test load.

3.5.2 Hoppers. The hopper shall have a capacity of not less than 50 pounds of flour when tested as specified in 4.4.4. A close-fitting cover equipped with handle shall be furnished. Hoppers and covers shall be fabricated of corrosion-resisting steel sheet specified in 3.4.1.

3.5.3 Screens. The sifting machine shall be provided with a sifting screen so supported and enclosed as to prevent the passage of unsifted flour. The cover for the screen-changing orifice shall be equipped with a soft gasket of solid synthetic rubber or felt permanently attached either to the cover or to the sifter body. A means for adjusting the compression on such gasket, when closing the orifice, shall be provided. The sifter screen shall be the manufacturer's standard commercial 14-mesh.

3.5.4 Casters. Casters shall be of the swivel type with semisteel wheels and each caster shall have a rated capacity of not less than 125 pounds. At least two of the casters on each machine shall be equipped with a locking device. The wheels shall be not less than 2 inches in diameter.

3.5.5 Motors. Unless otherwise specified (see 6.2), the motors will operate on a nominal system voltage of 120-volt, single phase, 60-hertz source. The motors shall be at least 1/4 horsepower rating and shall conform to CC-M-636. The motor housing may be an integral part of the sifter.

3.5.5.1 Cord. The machines shall be equipped with a 6-foot minimum length of cord. The cord shall be a 3-wire grounding, rubber-covered, flexible type, connected to the motor leads at one end and to a molded rubber 3-prong ground type attachment plug at the other end. The cord shall be reinforced at the motor end.

3.5.6 Electromagnetic compatibility. When specified (see 6.2), the flour sifter shall be designed and equipped for electromagnetic compatability in accordance with IIB of MIL-STD-461.

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3.5.7 Lubricating chart. The supplier shall affix to each machine a corrosion-resistant metal lubricating chart indicating all points of the equipment requiring lubrication, grade of lubricant required, and time intervals between lubrication.

3.5.8 Moisture and fungus resistant treatment. Unless otherwise specified (see 6.2), electrical components of equipment covered by this specification shall be moisture and fungus resistant treated with material conforming to and applied as required by MIL-T-152.

3.6 Finish. The exterior surfaces of the sifting machine shall be finished in accordance with the manufacturer's commercial practice.

3.7 Repair and maintenance. Provisions shall be made for ease of adjustment, service, and replacement of screens, motors, switches, and hoppers of the flour sifters. In addition, there shall be free access to inspect, service, and adjust any other component or equipment as provided by this specification, or as indicated by the component or equipment manufacturer without requiring disassembly of another component of equipment of the flour sifting machine.

3.8 Marking for identification.

3.8.1 Military agencies. Machines shall be marked in accordance with MIL-STD-130. Name plate shall be fastened in a readily accessible location.

3.8.2 Civil agencies. Unless otherwise specified (see 6.2), the manufacturer's commercial name plate will be acceptable.

3.8.3 Electrical characteristics. Electrical characteristics shall be indented, etched, or embossed on a corrosion-resistant metal plate, below the identification plate.

3.9 Workmanship.

3.9.1 Fabrication. Metal shall be free from fractures, splits, punctures, dents, and shall not be bowed or malformed. All burrs, slivers, or splinters shall be removed.

3.9.2 Welding or brazing. All surfaces of parts to be welded or brazed shall be clean. Welding and brazing shall be accomplished in a manner which will prevent the occurrence of burn holes, cracks, fracture, undercuts, or incomplete fusion. All scale or flux deposits shall be removed from finished welds. Welds shall be smooth and uniform.

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3.9.3 Rivets. Rivet shall be concentric with the hole and the head shall be in full contact with metal.

3.9.4 Threads. Threads shall not be missing, broken, stripped, or fractured.

3.9.5 Threaded fasteners. Threaded fasteners shall not be loose or missing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. When a preproduction sample is required, it shall be examined for the defects listed in table I, specified dimensions, and tested as specified in 4.4.1 through 4.4.5. Failure to pass any test shall be cause for rejection of the preproduction sample. The presence of any defect listed in table I or any dimension not within specified requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection. Sampling for inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Components and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, standards, and drawings unless excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Process examination. Examination shall be made that moisture and fungus treatment is applied as specified in 3.5.8 (when applicable). Whenever deviation is noted, correction shall be made to the affected item and lot in process.

4.3.3 End item inspection. The lot shall consist of all sifting machines offered for inspection at one time. The sample unit shall be one completely fabricated sifting machine.

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4.3.3.1 Visual examination. Examination shall be made of the flour sifters for the defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Finish not as specified	X	
Construction and workmanship general (applicable to all components and assemblies)	Blistered, pealed, chipped, or area of no film.	X	
	Fractured, split, punctured, dented, bowed, or malformed.	X	
	Sharp burr or edge, sliver, or splinter.	X	
	Any adjustable assembly that cannot be adjusted to perform the function intended.	X	
	Not connected or joined as specified.	X	
	Any functional component inoperative or will not function as intended.	X	
Gasket	Missing, not type specified.	X	
	Adjustment of gasket compression not provided.		X
Casters	One or more missing.	X	
	Not swivel type.	X	
	At least two not equipped with locking device.	X	
Stand	Step missing.	X	
	Stand missing.	X	
Bolt holes	Bolt holes in base of machine and on stand missing.	X	
Hopper	Cover missing.	X	
	Handle for hopper cover missing.		X
Screen	Missing.	X	

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TABLE I. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Rivets (when required)	Not concentric with hole.		X
	Head not in full contact with member.		X
Welding and brazing (where required)	Missing, incomplete fusion, burn holes, cracks fractures, or not otherwise fused.	X	
	Slag inclusion undercut not smooth and uniform, scale or flux deposits not removed.	X	
Threads	Missing, broken, stripped, or fractured.	X	
Threaded fasteners	Missing, loose, broken.	X	
Lubrication	Instruction missing.		X
Repair and maintenance	No provision for adjustment, service or replacement of screens, motors, switches, or hoppers.		X
Marking identification	Missing, incomplete, not legible, not proper location.		X

4.3.3.2 Dimensional examination. Inspection shall be made of the end item for compliance with dimensions specified herein. Any nonconformance with specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.3.3 Testing of the end item. When a preproduction sample is not required, the first unit of production shall be tested as specified in 4.4.1, 4.4.3, 4.4.4, and when applicable, 4.4.5. Each unit of production shall be tested as specified in 4.4.2. Any nonconformance shall be cause for rejection.

4.3.4 Examination of preparation for delivery. Examination of preservation, packaging, packing, and marking shall be in accordance with MIL-B-2316.

4.3.5 Standards compliance. Proof of compliance with the requirements of 3.3 shall be submitted to the Government representative.

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4.4. Tests.

4.4.1 Sifting. Fill the hopper with not less than 40 pounds of white flour conforming to type I, class A or B of N-F-481. Operate the sifting machine using a 14-mesh screen, and sift the flour. Measure the time required to determine compliance with the 40-pound per minute flour sifting capacity of 3.5. Any nonconformance with specified requirements shall constitute failure of this test.

4.4.2 Operational test. The sifting machine shall be connected to power and operated without load for a minimum of 5 minutes. While operating, observe bearings for overheating, and moving components for binding or improper operation, to determine compliance with the operating requirements of 3.5. Any nonconformance with specified requirements shall constitute failure of this test.

4.4.3 Step load. A 400 pound weight shall be placed so as to be concentrated within a space measuring 12 by 12 inches concentric with the geometric center of the step tread to determine compliance with the weight support and nontilting or nonshifting requirements of 3.5.1. Any noncompliance with specified requirements shall constitute failure of this test.

4.4.4 Capacity. Fill the hopper with flour; remove the flour; and weigh, to determine compliance with hopper capacity requirement of 3.5.2. Any nonconformance with specified requirements shall constitute failure of this test.

4.4.5 Electromagnetic compatibility tests. When electromagnetic compatability is required the preproduction sample or initial production unit, as applicable shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-461.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing for level A, B, or C shall be in accordance with MIL-B-2316.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

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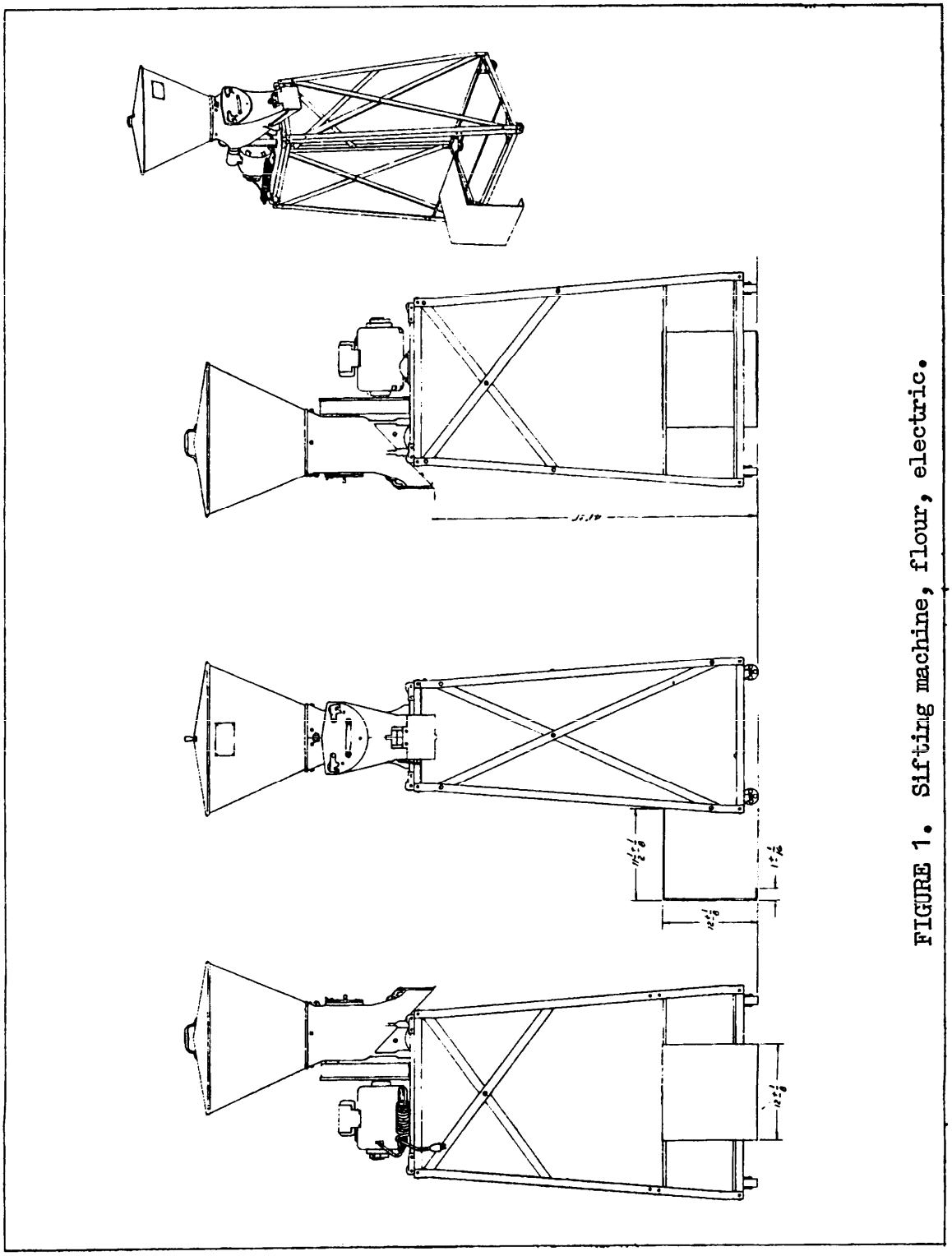


FIGURE 1. Sifting machine, flour, electric.

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5.2.2 Military agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The flour sifting machine covered by this specification is intended for use in Central Pastry Kitchens. Machines are used for resifting dry ingredients to remove foreign matter; break up lumps of flour, aerating; to cause more absorption of liquid for easier and faster development of dough and batters.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) When a first article is required (see 3.1, 4.2, and 6.3).
(c) Electric motor if other than specified (see 3.5.5).
(d) When electromagnetic compatability is required (see 3.5.6).
(e) When moisture and fungus resistance is not required (see 3.5.8).
(f) Identification marking for civil agencies, if other than specified (see 3.8.2).
(g) Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.2.1 Contract data requirements. Any requirements for equipment manuals for the item covered by this specification should be included in DD Form 1423 Contract Data Requirements List, and cited in the contract.

6.3 First article. When a first article is required it shall be tested and approved under the appropriate provisions of Armed Services Procurement Regulation 7-104.55. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test, and approval of the first article.

MILITARY CUSTODIANS:

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Preparing activity:

Army - GL

Project No. 7320-0275

Review activity:

Army - MD

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Sifting Machine, Flour, Electric 00-S-355B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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